

Liver Disease Prediction

Binary Logistic Regression in R — Model Output & Diagnostics

Explanatory logistic regression (glm, binomial family) modelling the probability of liver disease from clinical blood markers. The dataset contains 570 unique patients after removing duplicates. Below are the fitted model, its overall significance, effect sizes (odds ratios), and the standard regression assumption checks.

1. Fitted Model — summary(LiverModel.1)

Coefficients on the log-odds scale. DB, TP and ALB are statistically significant predictors ($p < 0.05$); ALB has a protective (negative) effect.

```
Call:
glm(formula = Selector ~ TB + DB + Alkphos + Sgpt + Sgot + TP +
    ALB, family = binomial(), data = LiverData)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.6211930   0.6554442  -0.948   0.34326
TB           0.0137504   0.1041096   0.132   0.89492
DB           0.5524006   0.2788175   1.981   0.04757 *
Alkphos      0.0011077   0.0007598   1.458   0.14489
Sgpt         0.0094342   0.0048873   1.930   0.05356 .
Sgot         0.0029389   0.0031517   0.932   0.35109
TP           0.3640501   0.1773259   2.053   0.04007 *
ALB          -0.6632389   0.2556939  -2.594   0.00949 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 684.11  on 569  degrees of freedom
Residual deviance: 569.60  on 562  degrees of freedom
AIC: 585.6

Number of Fisher Scoring iterations: 7
```

2. Overall Model Significance (Likelihood-Ratio Chi-Square)

The model explains the outcome significantly better than an intercept-only model: chi-square = 114.51 on 7 df, $p < 0.001$.

```
> modelChi <- LiverModel.1$null.deviance - LiverModel.1$deviance
> modelChi
[1] 114.5117
> chidf <- LiverModel.1$df.null - LiverModel.1$df.residual
> chidf
[1] 7
> chisq.prob <- 1 - pchisq(modelChi, chidf)
> chisq.prob
[1] 0
```

3. Odds Ratios & 95% Confidence Intervals

Exponentiated coefficients give odds ratios. DB (OR ≈ 1.74) and TP (OR ≈ 1.44) increase the odds of disease; ALB (OR ≈ 0.52) lowers them. Confidence intervals that exclude 1 confirm significance.

```
> LiverModel.1$coefficients
(Intercept)          TB          DB      Alkphos          Sgpt          Sgot
-0.621192959  0.013750448  0.552400590  0.001107714  0.009434218  0.002938912
          TP          ALB
  0.364050104 -0.663238907

> exp(LiverModel.1$coefficients)
(Intercept)          TB          DB      Alkphos          Sgpt          Sgot
  0.5373031  1.0138454  1.7374188  1.0011083  1.0094789  1.0029432
          TP          ALB
  1.4391463  0.5151800

> exp(confint(LiverModel.1))
Waiting for profiling to be done...
              2.5 %      97.5 %
(Intercept) 0.1475251 1.9439664
TB           0.9275245 1.5951998
DB           0.8380706 2.8943840
Alkphos      0.9997962 1.0027864
Sgpt         1.0006256 1.0199468
Sgot         0.9973740 1.0097045
TP           1.0188368 2.0447592
ALB          0.3100633 0.8464802
```

4. Multicollinearity — Variance Inflation Factor (VIF)

All VIF values are below 5 (max ≈ 3.9 for TP), so multicollinearity is not a concern. $1/\text{VIF}$ (tolerance) confirms this.

```
> vif(LiverModel.1)
          TB          DB      Alkphos          Sgpt          Sgot          TP          ALB
2.171699 2.308712 1.111823 1.770026 1.761629 3.763067 3.897670

> 1/vif(LiverModel.1)
          TB          DB      Alkphos          Sgpt          Sgot          TP          ALB
0.4604690 0.4331419 0.8994233 0.5649633 0.5676563 0.2657407 0.2565635
```

5. Correlation Matrix of Numeric Features

Pearson correlations (result\$r) and their p-values (result\$P). Note the strong pairwise correlations TB–DB (0.87), Sgpt–Sgot (0.79) and TP–ALB (0.78).

```
> num_data <- LiverData[sapply(LiverData, is.numeric)]
> result <- rcorr(as.matrix(num_data))
> result$r      # correlations
          Age          TB          DB      Alkphos          Sgpt
Age      1.000000000  0.011500166  0.0070496346  0.08167251 -0.08338338
TB        0.011500166  1.000000000  0.8741156676  0.20623926  0.21747068
DB        0.007049635  0.874115668  1.0000000000  0.23460937  0.23745043
Alkphos   0.081672511  0.206239259  0.2346093658  1.00000000  0.12682976
Sgpt     -0.083383380  0.217470681  0.2374504330  0.12682976  1.00000000
Sgot     -0.016753303  0.238677737  0.2584893568  0.16722982  0.79185718
TP       -0.197051624 -0.008588269 -0.0008750039 -0.03004849 -0.03519349
ALB      -0.271169714 -0.224124439 -0.2307513669 -0.16831764 -0.02797280
A.G.Ratio -0.215506271 -0.207473516 -0.2012275157 -0.23584767 -0.00639423
```

```

      Sgot      TP      ALB      A.G.Ratio
Age      -0.01675330 -0.1970516239 -0.27116971 -0.21550627
TB        0.23867774 -0.0085882686 -0.22412444 -0.20747352
DB        0.25848936 -0.0008750039 -0.23075137 -0.20122752
Alkphos    0.16722982 -0.0300484933 -0.16831764 -0.23584767
Sgpt       0.79185718 -0.0351934945 -0.02797280 -0.00639423
Sgot       1.00000000 -0.0219996289 -0.08518048 -0.07278262
TP        -0.02199963  1.0000000000  0.78473136  0.23350330
ALB       -0.08518048  0.7847313644  1.00000000  0.68385426
A.G.Ratio -0.07278262  0.2335033024  0.68385426  1.00000000

> result$P # p-values
      Age      TB      DB      Alkphos      Sgpt
Age      NA 7.841084e-01 8.666302e-01 5.130966e-02 4.660778e-02
TB      7.841084e-01 NA 0.000000e+00 6.814494e-07 1.575360e-07
DB      8.666302e-01 0.000000e+00 NA 1.442843e-08 9.530234e-09
Alkphos  5.130966e-02 6.814494e-07 1.442843e-08 NA 2.416297e-03
Sgpt     4.660778e-02 1.575360e-07 9.530234e-09 2.416297e-03 NA
Sgot     6.897977e-01 7.953939e-09 3.738376e-10 6.018625e-05 0.000000e+00
TP       2.129022e-06 8.378880e-01 9.833696e-01 4.740007e-01 4.016666e-01
ALB      4.591105e-11 6.369731e-08 2.512623e-08 5.378141e-05 5.050879e-01
A.G.Ratio 2.047066e-07 5.824063e-07 1.276851e-06 1.205029e-08 8.789294e-01
      Sgot      TP      ALB      A.G.Ratio
Age      6.897977e-01 2.129022e-06 4.591105e-11 2.047066e-07
TB       7.953939e-09 8.378880e-01 6.369731e-08 5.824063e-07
DB       3.738376e-10 9.833696e-01 2.512623e-08 1.276851e-06
Alkphos  6.018625e-05 4.740007e-01 5.378141e-05 1.205029e-08
Sgpt     0.000000e+00 4.016666e-01 5.050879e-01 8.789294e-01
Sgot     NA 6.001781e-01 4.206386e-02 8.253733e-02
TP       6.001781e-01 NA 0.000000e+00 1.693239e-08
ALB      4.206386e-02 0.000000e+00 NA 0.000000e+00
A.G.Ratio 8.253733e-02 1.693239e-08 0.000000e+00 NA

```

6. Linearity of the Logit (Box-Tidwell Test)

Each continuous predictor is multiplied by its log and added to the model. None of the interaction terms (logTB ... logALB) is significant, so the linearity-of-the-logit assumption holds.

```

> LiverData$logTB <- log(LiverData$TB) * LiverData$TB
> LiverData$logDB <- log(LiverData$DB) * LiverData$DB
> LiverData$logAlkphos <- log(LiverData$Alkphos) * LiverData$Alkphos
> LiverData$logSgpt <- log(LiverData$Sgpt) * LiverData$Sgpt
> LiverData$logSgot <- log(LiverData$Sgot) * LiverData$Sgot
> LiverData$logTP <- log(LiverData$TP) * LiverData$TP
> LiverData$logALB <- log(LiverData$ALB) * LiverData$ALB
>
> LiverTest.1 <- glm(Selector ~ TB + DB + Alkphos + Sgpt + Sgot + TP + ALB
+ logTB + logDB + logAlkphos + logSgpt
+ logSgot + logTP + logALB,
+ data = LiverData, family = binomial())
Warning message:
glm.fit: fitted probabilities numerically 0 or 1 occurred
>
> summary(LiverTest.1)

Call:
glm(formula = Selector ~ TB + DB + Alkphos + Sgpt + Sgot + TP +
ALB + logTB + logDB + logAlkphos + logSgpt + logSgot + logTP +
logALB, family = binomial(), data = LiverData)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.5500814   4.4910929   0.790   0.429
TB          -0.2185576   0.7765495  -0.281   0.778
DB           1.0318002   1.0949233   0.942   0.346
Alkphos      0.0108270   0.0094970   1.140   0.254
Sgpt         0.0297342   0.0221047   1.345   0.179

```

Sgot	-0.0015156	0.0263770	-0.057	0.954
TP	-3.6092628	2.7652129	-1.305	0.192
ALB	2.0993022	2.3205588	0.905	0.366
logTB	0.0528611	0.1768242	0.299	0.765
logDB	-0.1964960	0.3350623	-0.586	0.558
logAlkphos	-0.0013332	0.0012862	-1.037	0.300
logSgpt	-0.0039036	0.0036084	-1.082	0.279
logSgot	0.0007709	0.0045215	0.170	0.865
logTP	1.3862749	0.9696619	1.430	0.153
logALB	-1.2668662	1.0652582	-1.189	0.234

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 684.11 on 569 degrees of freedom
Residual deviance: 565.73 on 555 degrees of freedom
AIC: 595.73

Number of Fisher Scoring iterations: 9